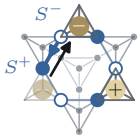


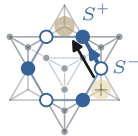
i.a) create


 $S_i^+ S_j^-$: pair splits

i.b) move


 $S_i^+ S_j^-$: spinon walks

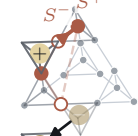
i.c) annihilate


 $S_i^+ S_j^-$: + meets -
 $\mathbf{q}_\gamma = (0, 0, 0)$

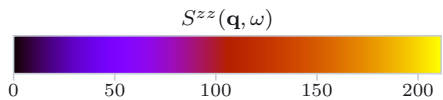
i.d) create


 $S_i^+ S_j^-$: pair splits

i.e) separate


 $S_i^+ S_j^-$ moves the -

i.f) annihilate


boundary maps - onto +
 $\mathbf{q}_\gamma = (0, 0, -1)$


iii.a) periodic ED



iii.b) winding-free

